

Statistical Structure of Charge Disorder in Si/SiGe Quantum Dots

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Properties of quantum dot based spin qubits have significant inter-device variability due to unavoidable presence of various types of disorder in semiconductor nanostructures. A significant source of this variability is charge disorder at the semiconductor-oxide interface, which causes unpredictable, yet, as we show here, correlated fluctuations in such essential properties of quantum dots like their mutual tunnel couplings, and electronic confinement energies. This study presents a systematic approach to characterize and mitigate the effects of such disorder. We utilize finite element modeling of a Si/SiGe double quantum dot to generate a large statistical ensemble of devices, simulating the impact of trapped interface charges. This work results in a predictive statistical model capable of generating realistic artificial data for training machine learning algorithms. By applying Principal Component Analysis to this dataset, we identify the dominant modes through which disorder affects the multi-dimensional parameter space of the device. Our findings show that the parameter variations are not arbitrary, but are concentrated along a few principal axes – i.e. there are significant correlations between many properties of the devices. We finally compare that against control modes generated by sweeping the gate voltages, revealing limitations of the plunger-only control. This work provides a framework for enhancing the controllability and tunability of spin qubit devices, by systematically addressing the nature of electrostatic disorder that leads to statistical correlations in properties of double quantum dots.

I. INTRODUCTION

Spin qubit devices based on quantum dots in Si/SiGe nanostructures already offer high-fidelity single-qubit and two-qubit gates [1–8], and readout [9, 10], while maintaining relatively long coherence times [2, 11], and shuttling-based connectivity [12–17]. This makes them a viable platform for both digital [18, 19] and analog quantum computing [20, 21].

The key long-term advantage of quantum computing architecture based on silicon quantum dots is the possibility of leveraging industrial manufacturing technologies [22–25] to create devices containing millions of qubits on a chip that can easily fit into a standard dilution refrigerator. However, variability of properties of such qubits realized in a solid-state technology remains a challenge for quantum computing architecture [26], as much more attention has to be given to tuning of single-qubit properties, and parameters of two-qubit gates than in case of qubits based on identical atoms or ions [27, 28]. The presence of large variability of parameters of quantum dot devices created according to the same blueprint on the same heterostructure has been an experimental challenge since the beginning of research into quantum dot based spin qubits. However, only recently developments leveraging industrial technology have allowed for making measurements on large numbers of quantum dots defined on the same wafer [23], leading to a qualitative change in the amount of data on variability of properties of qubits

created in a given structure. Such developments provide motivation for calculations of statistical distributions of various parameters of qubits for given microscopic models of disorder in nanostructures.

Variability of properties of qubits in Si/SiGe quantum dots can be caused by electrostatic disorder due to the presence of charged impurities and defects at random positions in the nanostructure [29, 30], atomistic disorder and interface roughness at Si/SiGe interface [31–34] and inhomogeneous strain [35]. These effects were also discussed theoretically for SiMOS devices [31, 36], but the degree to which their conclusions from these works apply to Si/SiGe (for which the distance between the qubits and charge traps in an insulator, and the nature of interface roughness are distinct from SiMOS) is unclear.

Due to large variability of the basic parameters of devices hosting multiple quantum dots, such as on-site energies and interdot tunnel couplings, the voltages on the gates defining the multi-dot system have to be tuned in order to even start considering the system as a multi-qubit register. For registers consisting of $N > 2$ qubits this tuning quickly becomes unmanageable with increasing N if done manually, and it has to be automated [37, 38]. Machine learning algorithms used for these purposes have to be trained on simulated data on response of disordered multi-dot devices to control voltages [39]. Identifying the realistic microscopic model of disorder that captures the major features of qubit variability would allow then for generation of more realistic training data.

In this paper, we focus on electrostatic disorder due to density ρ of negative charges trapped at SiGe/SiO₂ inter-

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face, ranging from 5×10^9 to $5 \times 10^{10} \text{ cm}^{-2}$ [29, 30, 40], and consider the variability of parameters of a double quantum dot (DQD) caused by it. Using finite element modeling of a realistic Si/SiGe double quantum dot with trapped interface charges, we generate a statistical ensemble of devices. The DQD is controlled by two plunger gates, and also possibly by a barrier gate [41, 42]. At given ρ , for every realization of disorder, we tune the voltages on these gates to identify the range of voltage detunings at which a single electron is transferred between the two dots and, in this way, obtain a functional DQD device. The detuning needed to bring the DQD into a symmetric shape at which the charge transition occurs is the first quantity on the statistics of which we focus. We then reconstruct the statistical distribution of many other quantities: interdot tunnel coupling t_c , interdot distance d , interdot barrier height h_B , orbital energies and confinement lengths in both dots, and disorder-induced electric fields in the z direction acting on both QDs. Next we move beyond quantifying the magnitude of parameter variations, and uncover the characteristic ‘modes’ of disorder. We demonstrate that disorder modifies parameters of DQDs along specific directions in the parameter space, and certain parameters exhibit significant correlations. An interesting example is a strong correlation between interdot distance d and interdot barrier height h_B , the presence of which allows us to identify a robust quantitative relation between d and t_c .

To uncover and quantitatively analyze the characteristic modes of disorder, and to further assess their controllability, we employ Principal Component Analysis (PCA) [43], which is a dimensionality reduction method based on a spectral analysis of the covariance matrix. Within the spin qubit literature, the application of PCA has been largely limited to technical pre-processing steps, such as improving the signal-to-noise ratio of measurement data [44]. In contrast, here we use it to identify the principal directions in the parameter space along which fluctuations occur (the eigenvectors) and to quantify their corresponding variance (the eigenvalues). When the variance is concentrated in just a few dominant components, one can effectively truncate the parameter space and perform the analysis within a low-dimensional manifold [45]. In our case, this manifold captures the essential physics of the electrostatic disorder, and allows us to build a resource-efficient predictive statistical model of a disordered DQD using data from microscopic simulations. We have applied the PCA method to our simulated disorder data and found three statistically relevant modes that correspond to characteristic features of spatial realizations of charge disorder. The mode responsible for the most variability, for instance, is caused by fluctuations of number of charged defects located *between* the two dots. By comparing the manifold spanned by these disorder modes to the one spanned by the control modes, we found that the effects of this dominant mode cannot be reversed without the use of the barrier gate. This framework thus provides a clear, quantitative demonstration

of the limitations of plunger-only control scheme [46], in mitigating realistic device-to-device variability. Our work establishes thus PCA as a primary framework for physical interpretation, using it to deconstruct the structure of device-to-device variability for the microscopic model of disorder discussed here, and to rigorously quantify system controllability.

The paper is structured as follows. Section II introduces our device model. We then quantify the device-to-device variability in Sec. III, first by its impact on functional device yield (defined by thresholds for amount of correction to detuning needed to obtain a symmetric DQD, and values of orbital energies and tunnel couplings of the resulting device), and then by analyzing the underlying fluctuations in tunnel coupling. The tunability of tunnel coupling in presence of disorder, which has crucial significance for turning on and off the two-qubit interactions, is discussed in Sec. III D. To find the root causes of this variability, in Sec. IV we use the principal component analysis to identify the dominant disorder modes, and build a predictive statistical model of disorder. This framework is then applied in Sec. V to quantify the effectiveness of gate control in mitigating these effects, comparing standard two-gate and three-gate schemes. We conclude with a discussion in Sec. VI. Appendix A details the supplementary results of PCA and DQD parameter fluctuations, while Appendix B provides the results for controllability, all relating to the alternative operational gate voltage setting. Finally, Appendix C presents the correction recipe and shows, for individual disorder realizations, how the PCA-based voltage update restores the device toward the symmetric operating point.

II. DEVICE MODEL AND SIMULATION

We model the double quantum dot using the finite element method, as implemented in COMSOL Multiphysics [47]. This allows us to determine the electrostatic confinement potential used in the single-electron Schrödinger equation. The model, illustrated in Fig. 1, includes metallic top gates and spatially random distribution of fixed charges at the SiO_2/Si cap interface having density ρ . Our simulated heterostructure is grown along the $\hat{z}$ axis corresponding to [001] crystallographic direction, and consists of a Si quantum well between two SiGe barriers, the Si cap, two insulating regions (SiO_2), and a screening region. Metallic gates are employed to trap and confine electrons within the Si layer, and to induce an external electric field along the $\hat{z}$ direction, thereby enabling tunability of the electronic states.

For concreteness, we analyze a quantum well of thickness $h_{\text{Si}} = 10 \text{ nm}$ [11, 48]. However, it should be noted that with the applied electric field in z direction $F_z = 5 \text{ MV/m}$ the electron wavefunction is localized within $\approx 5 \text{ nm}$ from the top Si/SiGe interface, and the exact value of h_{Si} that is larger than this localization length is irrelevant when considering the effects of electrostatic disorder

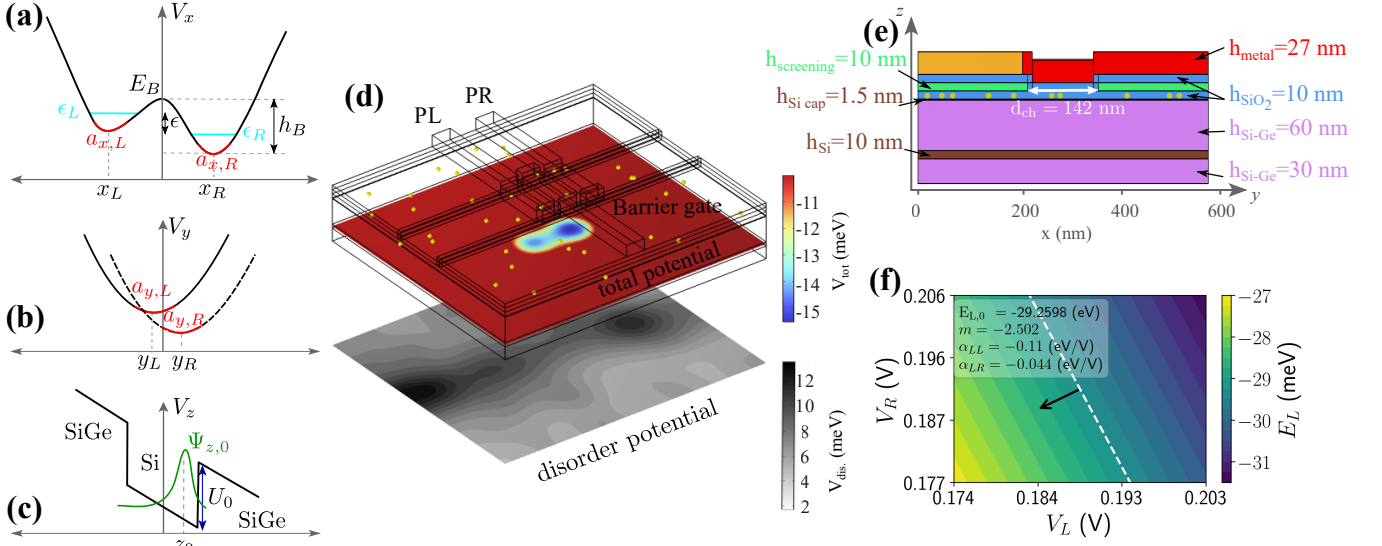


FIG. 1. **Device model and experimental regime.** (a)-(c) schematic diagrams of double-well potential along three axes used to present relevant DQD parameters (d) Schematic of the simulated Si/SiGe DQD device structure showing the gate layout and trapped charges. Example distribution of the disorder potential ($V_{dis.}$) generated solely by negatively charge defects ($\rho = 10^{10} \text{ cm}^{-2}$), shown with all gates turned off. (e) side view of DQD device (f) Extracted plunger gate lever arm α , confirming agreement with typical experimental values.

on electronic states. Hence our calculations apply also to the case of thinner quantum wells, such as those with $h_{\text{Si}} = 5 \text{ nm}$ chosen to be below the Matthews-Blakeslee critical thickness for strain relaxation [44, 49].

The interface between the upper SiGe barrier and the insulating region is at $h_{\text{SiGe}}^{\text{top}} = 60 \text{ nm}$, distance from the Si/SiGe interface, and it is followed by an additional 1.5 nm of Si cap layer. We use a relative permittivity of $\epsilon_r = 3.9$ for the SiO_2 oxide and $\epsilon_{\text{Si}} = 12$, $\epsilon_{\text{SiGe}} = 13.2$. Following prior Si/SiGe quantum-dot modeling, we take a conduction-band offset of $U_0 = 150 \text{ meV}$ for Si well with $\text{Si}_{0.7}\text{Ge}_{0.3}$ barriers [50]. In the insulating region, we impose $U \rightarrow \infty$ (or a hard-wall condition in the COMSOL solver) to ensure electron confinement. The electrostatic potential is obtained by solving Poisson's equation with homogeneous Neumann boundary conditions on the outer domain boundaries, while the screening metal, introduced to capture the screening effect and the resulting mirror charges, is maintained at a constant Dirichlet reference potential ($V_{\text{scr}} = 0$). Additional test simulations show that the electric field distribution in the region of the quantum well in which the DQD is formed is weakly dependent on the height of the bottom SiGe buffer, and in the simulations described here we use $h_{\text{SiGe}}^{\text{bottom}} = 30 \text{ nm}$, see Fig. 1(e). To the solutions for electrostatic potential obtained with these boundary conditions we add a potential given by $-eF_z z$, and in this way we include the presence of vertical electric field due to voltage difference between the top gates (modeled explicitly in our structure) and a back gate (absent in the model of structure from Fig. 1(e)) that is typically at least a micron below the quantum well. This framework captures the essential physics of electron confinement in

Si/SiGe quantum dots, including the anisotropy of the effective masses, gate-induced fields, lateral ellipticity, heterostructure band offsets, and the presence of F_z electric field in the accumulation-mode heterostructure.

The device structure shown in Fig. 1 (d) has overall dimensions of $660 \times 582 \text{ nm}^2$. The channel is defined by a gap in the screening layer, with a width of $d_{\text{channel}} = 142 \text{ nm}$. The metallic gates have a width of $w_{\text{metal}} = 45 \text{ nm}$ and a height of $h_{\text{metal}} = 27 \text{ nm}$. The remaining dimensions are provided in Fig. 1(e). We approximate the effectively two-dimensional potential that leads to lateral confinement by evaluating the numerically obtained three-dimensional potential $V(x, y, z)$ at z corresponding to expectation value of this coordinate for an electron confined in the quantum well. The numerically obtained lateral confinement is then fitted near the two potential minima by harmonic potentials characterized by $a_{x(y)} = m_t \omega_{x(y)}^2$ curvatures, where $\hbar \omega_{x(y)}$ are the energies of quantum harmonic oscillator, $m_t = 0.19 m_e$ is the relevant effective mass, and the resulting characteristic in-plane confinement lengths are $L_{x(y)} = (\hbar^2 / m_t a_{x(y)})^{1/4}$.

In the absence of charged defects, we tune the device to achieve a tunneling gap of $2t_{c,0} \approx 24 \mu\text{eV}$ and an interdot distance of $d_0 = 95 \text{ nm}$, corresponding to a slightly elliptical quantum dot with lateral dimensions $L_{x,0} = 20 \text{ nm}$ and $L_{y,0} = 18 \text{ nm}$, giving a ratio of $r_{L/R} = L_{x,0} / L_{y,0} \approx 1.08$. These parameters are both realistic and experimentally relevant [51, 52]. The energy contribution from the applied vertical electric field is described by the term $-eF_z z$, where the field strength in defect-free DQD is $F_{z,0} = 5.34 \text{ MV/m}$. To achieve this, we set plunger gates $V_{L/R} = 0.35 \text{ V}$ to obtain an orbital

excitation energy of $E_{\text{orb}} \approx 1.5$ meV in each dot, with the barrier gate set to $V_B = -0.1$ V.

These values have been chosen to guarantee that in presence of $\rho = 5 \times 10^{10} \text{ cm}^{-2}$ charges at the semiconductor/oxide interface the electrostatic potential in the Si quantum well still possesses two local minima, i.e. the DQD can be still found in the presence of disorder. In such a gate-defined quantum dot, the energy at the left potential minimum, E_L , can be written to a very good approximation as a linear function of plunger voltages,

$$E_L(V_L, V_R) = \alpha_{LL}V_L + \alpha_{LR}V_R + c_L, \quad (1)$$

and we determine the lever arms α_{LL}, α_{LR} by ordinary least squares fit to the calculated (V_L, V_R, E_L) data. Lever arms quantify how efficiently each physical gate shifts the dot energy—information used to construct virtual gates [53–55] that independently control detuning and occupancy while minimizing cross-talk, thereby enabling precise qubit calibration. The constant-energy contours (charge transition lines) in the (V_L, V_R) plane satisfy $E_L = \text{const}$ and thus have slope $m = -\alpha_{LL}/\alpha_{LR}$. The energy gradient line in gate space is orthogonal to the contours and indicating the gate combination that most strongly shifts E_L . Extracted lever-arm parameters, defining constant-energy contours and energy gradients in (V_L, V_R) space for the defect-free device, are shown in Fig. 1(f). Our analysis shows that the charge disorder induced variation in the lever arm remains below 10% across the considered density range. As a result, we have chosen to exclude it from further investigations.

To characterize the DQD in presence of a given spatial realization of disorder (positions of charges at the interface), we first set the gate voltages to the previously optimized values, and then tune the plunger gates to locate the tunnel anticrossing. The observation of this anticrossing confirms the formation of a DQD. Once the anticrossing is identified, the in-plane potential is extracted and fitted to its two minima, as illustrated in Fig. 1 (a). We begin by fitting the most relevant parameters of the double-well potential— E_D (the minimum of the parabola for dot D), the dot position (x_D, y_D) , and the curvatures $a_{x(y),D}$. The single-dot lateral confinement energy is $\epsilon_D = E_D + \hbar(\omega_{x,D} + \omega_{y,D})/2$. The value of the electric field $F_{z,D}$, with or without impurities, is obtained at each dot position D .

In presence of interface charges the two minima of the potential obtained for initial plunger voltages are detuned. We obtain this detuning by finding the voltage difference that makes the DQD symmetric, where, we adjust the difference between the two plunger potentials while keeping their average and the barrier gate voltage constant, and using the lever arm to convert it into the disorder-induced energy detuning: In presence of interface charges the two minima of the potential obtained for initial plunger voltages are detuned. We obtain this detuning by finding the voltage difference that makes the DQD symmetric, where, we adjust the difference between the two plunger potentials while keeping their average

and the barrier gate voltage constant, and using the lever arm to convert it into the disorder-induced energy detuning:

$$\epsilon = \alpha \frac{\Delta V_{LR}}{2}, \quad \alpha = \alpha_{LL} + \alpha_{RR} - \alpha_{LR} - \alpha_{RL}. \quad (2)$$

For the disorder-free case where the value of $|\alpha|$ is 0.12 eV/V. In presence of $\rho \sim 10^{10} \text{ cm}^{-2}$ defect charges the variation of α is $< 10\%$.

Among the many parameters involved, we focus on a subset that mainly governs DQD qubit operations. This parameter vector, which forms the basis of our statistical analysis, includes the tunnel coupling t_c (tunneling gap $2t_c$), the interdot distance d , the interdot barrier height h_B , the disorder-induced energy detuning ϵ , the average confinement length L_x along the axis connecting the two dots and its left–right difference ΔL_x , and the average vertical electric field F_z together with its left–right difference ΔF_z , and the confinement shape ratio $r_D = L_x/L_y$ at dot D .

III. QUANTIFYING DEVICE-TO-DEVICE VARIABILITY

In this section, we quantify the variability of the double quantum dot by analyzing an ensemble of devices subject to independent realizations of charge disorder. This model aims to reconstruct device-to-device, or thermal-cycle variability observed in the experiments [39]. For each instance of disorder, we tune the dots to zero energy detuning before extracting their final parameters. This ensures we are analyzing variability around a consistent operational point.

A. Impact on Tunability

While the fluctuations of individual parameters are important, the ultimate test of scalability is how many devices in an array are functional. A single critical parameter falling out of its operational range can lead to unusable qubit. To capture this, we define the *tunability success rate (TSR)* as the percentage of DQD instances that meet specific, experimentally motivated criteria after pre-tuning.

For a DQD to be considered tunable, we demand that: (i) the orbital energy in each dot remains above 1 meV to suppress thermal excitations [56]; (ii) the tunnel coupling lies in the range of 10 – 250 μeV , which is required for high-fidelity gate operations [57, 58]; (iii) the barrier height above 0.5 meV to preserve DQD potential [59], and (iv) the initial detuning can be corrected with less than 20 mV of plunger voltage, limiting the required tuning range [60].

Our analysis reveals that the TSR is critically affected by the charge density ρ . As shown in Fig. 2, the TSR decreases sharply with increasing disorder. For a more

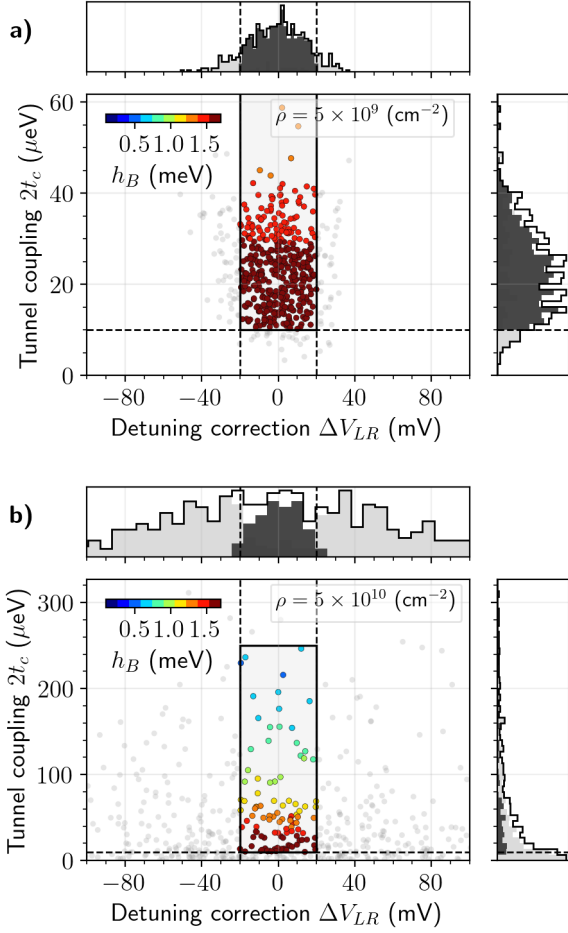


FIG. 2. Estimated tunability success rates of 76% and 20% are obtained for (a) $\rho = 5 \times 10^9 \text{ cm}^{-2}$ and (b) $\rho = 5 \times 10^{10} \text{ cm}^{-2}$, respectively. The results are obtained under constraints of a plunger gate correction below 20 mV, orbital energies above 1 meV in both dots, and tunnel gaps ranging between 10 and 250 μeV . The histograms in the top and right panels show the distributions of tunnel coupling and detuning correction, respectively. The dark (light) gray shading indicates the number of samples that fall inside (outside) the success box. The overlay step histogram showing the entire distribution, including all the data points.

quantitative comparison, Figs. 2 (a) and (b) show realizations of parameter samples for $\rho = 5 \times 10^9 \text{ cm}^{-2}$ and $\rho = 5 \times 10^{10} \text{ cm}^{-2}$. The side histograms display the distributions of tunnel coupling and detuning correction, while the color scale represents the barrier height h_B . We observe that at lower density, the estimated TSR is approximately 76%, whereas at higher density, $\rho = 5 \times 10^{10} \text{ cm}^{-2}$, it decreases to 20%.

B. Fluctuations in Key Operational Parameters

To quantify the contribution of each individual parameter to the overall variability, we now examine the

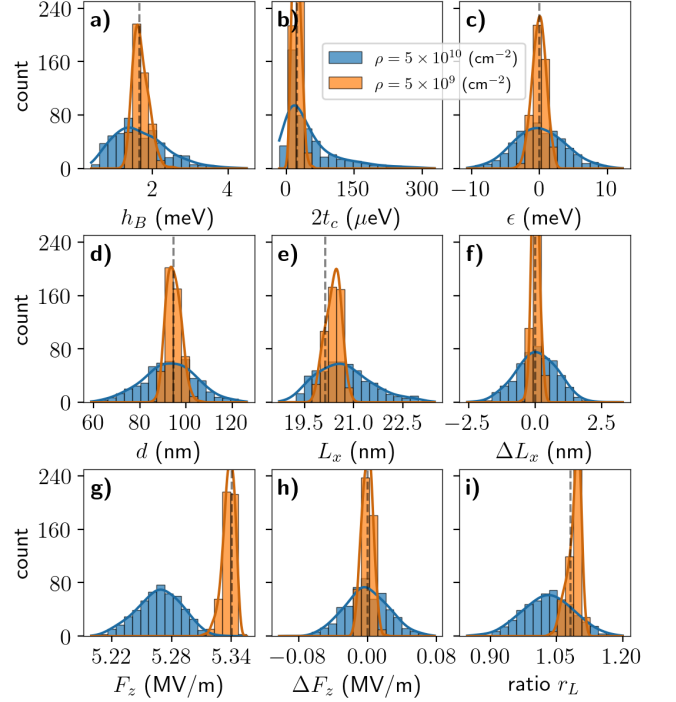


FIG. 3. Marginal distributions of the key DQD parameters as function of charge density ρ . The corresponding ensemble-averaged value of each DQD parameter are summarized in Table I. The vertical dashed line represents the defect-free DQD parameters, serving as a reference for a device where the gates are set to $V_{L/R} = 0.35 \text{ V}$ and $V_B = -0.1 \text{ V}$.

variability of each DQD parameter separately. Figure 3 shows the marginal distributions for key parameters at two charge densities. While many distributions are approximately Gaussian, parameters like the tunnel coupling (t_c), barrier height (h_B) and electric field (F_z) exhibit significant non-Gaussian tails, including a finite probability of extreme variations that can compromise device function.

As shown in Figure 3, an increase in charge density generally widens the parameter distributions. Higher densities increase the probability of smaller interdot distances d , creating a much heavier tail in the distribution for tunnel couplings t_c . As detailed in Table I, this leads to more than a threefold increase of the standard deviation of d , resulting in a six-fold increase of standard deviation of t_c .

Crucially, while shifts in mean values can often be compensated for by tuning gate voltages, we find that the relative variability, σ/μ , consistently increases with charge density for all parameters. This highlights a fundamental challenge, as the intrinsic randomness becomes more pronounced at higher disorder levels.

TABLE I. Statistical summary of parameters, including mean, standard deviation (std), and the coefficient of variation (CV) for two densities $\rho_1 = 5 \times 10^9 \text{ cm}^{-2}$ and $\rho_2 = 5 \times 10^{10} \text{ cm}^{-2}$.

Parameter		Mean (μ)	Std (σ)	CV ($\sigma/ \mu $)
d (nm)	ρ_1	94.60	3.10	0.033
	ρ_2	92.75	11.23	0.12
h_B (meV)	ρ_1	1.691	0.203	0.12
	ρ_2	1.678	0.692	0.41
ϵ (meV)	ρ_1	-0.072	0.987	—
	ρ_2	0.036	3.683	—
F_z (MV/m)	ρ_1	5.337	0.006	0.001
	ρ_2	5.267	0.022	0.004
ΔF_z (MV/m)	ρ_1	$< 10^{-4}$	0.006	—
	ρ_2	$< 10^{-3}$	0.024	—
ΔL_x (nm)	ρ_1	0.009	0.139	—
	ρ_2	0.049	0.761	—
L_x (nm)	ρ_1	20.37	0.235	0.012
	ρ_2	20.709	0.837	0.040
$2t_c$ (μeV)	ρ_1	23.13	9.13	0.40
	ρ_2	50.30	57.91	1.15
$r_L, [r_R]$	ρ_1	1.090, [1.090]	0.0146, [0.0144]	0.013, [0.013]
	ρ_2	1.030, [1.030]	0.0561, [0.0566]	0.054, [0.055]

Note. CV is not informative when the mean is close to zero or changes sign (e.g., ϵ , ΔF_z , ΔL_x); the large values reflect the instability of $\sigma/|\mu|$. For these quantities, variability is better characterized by σ . Means slightly offset from zero likely arise from finite-size and numerical-accuracy effects.

C. Tunnel Coupling Fluctuations

We now focus on the non-Gaussian parameter, tunnel coupling t_c , which is critical for the speed and fidelity of two-qubit gates [5]. According to the WKB approximation [61, 62], t_c depends exponentially on both the inter-dot distance d and barrier height h_B : $t_c = t_{c,0} e^{-\beta d \sqrt{2m h_B}/\hbar}$. Thus even small fluctuations in d and h_B can lead to large variations in t_c , as shown by the first-order approximation:

$$\delta t_c \approx -t_c \frac{\beta \sqrt{2m}}{\hbar} \left(\sqrt{h_B} \delta d + \frac{d}{2\sqrt{h_B}} \delta h_B \right). \quad (3)$$

Hence the variability in tunnel coupling, quantified as $\langle \delta t_c^2 \rangle$, depends on the variances of both d and h_B , as well as their covariance, $\langle \delta d \delta h_B \rangle$.

Figure 4 (a)-(b) shows the simulated tunnel coupling t_c as a function of inter-dot distance d for two charge disorder densities. For both densities, we observe a strong, non-linear correlation consistent with the WKB approximation, confirming that as disorder pushes the dots closer together, the tunnel coupling increases dramatically. The higher disorder density in panel (a) can push some devices into a small-separation ($d < 80$ nm) regime where the WKB approximation breaks down. However, the

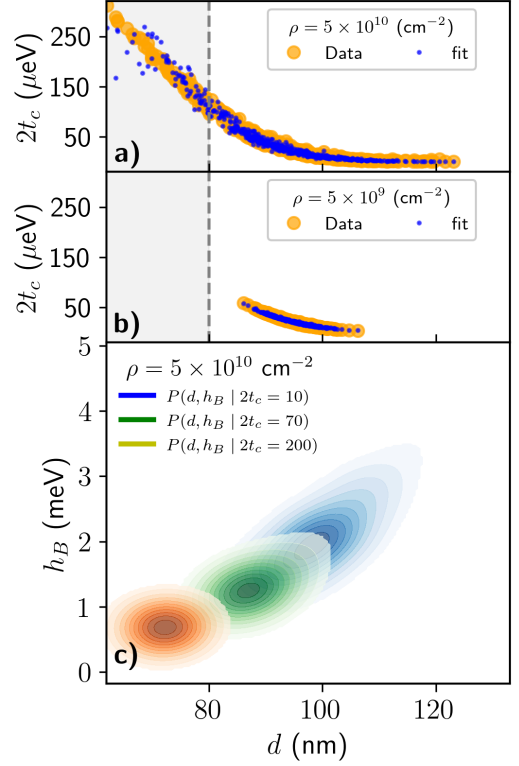


FIG. 4. Fluctuations in the tunnel coupling t_c as a function of interdot distance d for two charge densities: (a) $\rho = 5 \times 10^{10} \text{ cm}^{-2}$ and (b) $\rho = 5 \times 10^9 \text{ cm}^{-2}$. (c) Joint conditional density $P(d, h_B | 2t_c = \text{value})$ for three values of t_c . The results indicate that variations in t_c are primarily driven by fluctuations in d , which is positively correlated with h_B .

strong correlation between t_c and d is preserved. This suggests that the fluctuations in tunnel coupling can be primarily explained by the fluctuations in d , which is positively correlated with h_B as shown in Fig. 4 (c). Thus influence of h_B can be effectively absorbed into a renormalized coefficient β' in $\delta t_c \sim -t_c \frac{\beta'}{\hbar} \sqrt{h_B} \delta d$, and allows the barrier height variations to be excluded from our subsequent analysis of disorder modes.

The strong correlation between t_c and d indicates that measuring t_c fluctuations across an ensemble of devices can serve as a direct probe of variations in their physical separation. Furthermore, disorder-induced increases in d can lead to a severe suppression of the tunnel coupling, significantly impairing spin qubit operations. This issue is particularly detrimental for applications like bucket-brigade charge shuttling [63, 64], in which an electron is supposed to be adiabatically transferred by global voltage pulses along a chain of tunnel-coupled quantum dots. The probability of unsuccessful charge shuttling, and the amount of spin dephasing of the shuttled qubit, is expected to be dominated by the interdot transfer across weak links, i.e. pairs of adjacent QDs having particularly low values of t_c [64–66]. For example, for $\rho = 5 \times 10^9 \text{ cm}^{-2}$ ($5 \times 10^{10} \text{ cm}^{-2}$), we can observe a collapse of the

tunnel gap of $2t_c \leq 10 \mu\text{eV}$ in 6% (25%) of our simulated cases. The difference can be attributed to the large variance at larger densities, which on one hand increases the mean, but at the cost of an increasing probability of sampling an outlier. Dealing with such a weakly tunnel-coupled pair of quantum dots requires individual tuning that will make long distance shuttling resource intensive. In contrast, the relatively small variance gives an additional possibility of global retuning of the quantum dot towards typically shallower, more coupled dots, without inducing unwanted outliers. As demonstrated in Appendix A, at lower defect charge density $\rho = 5 \times 10^9 \text{ cm}^{-2}$ one can adjust the tuning point diminish the fraction of DQDs with $2t_c < 10 \mu\text{eV}$ to 2.5%, as an effect of moving the average to larger values, while keeping the variance in a reasonable range, which is impossible at larger densities.

D. Tuning of Tunnel Coupling with Barrier Gate in Presence of Disorder

When each of the spins in the two quantum dots is treated as a single-spin qubit, the ability to change the tunnel coupling by changing V_B is crucial for turning on and off the two-qubit exchange interaction that is used to perform two-qubit gates [3–8].

Our findings indicate that the sensitivity of the tunnel coupling to the barrier voltage, defined by the gradient $\partial \log_{10}(2t_c)/\partial V_B$, is remarkably robust to disorder and remains essentially independent of the charged defect density. In Fig. 5 we show the dependence of t_c on barrier voltage for ten realizations of disorder (five for each of two densities, $\rho_1 = 5 \times 10^9 \text{ cm}^{-2}$ and $\rho_2 = 5 \times 10^{10} \text{ cm}^{-2}$). We see there that for all disorder realizations and density levels (ρ_1, ρ_2) exhibit nearly identical slopes. However, higher disorder densities significantly increase the vertical spread of the initial t_c values. Because the sensitivity is constant, this increased absolute spread necessitates a much larger sweep of V_B to bring every device in an ensemble down to a common target (e.g., the $1 \mu\text{eV}$ threshold required for single-qubit operations). This demonstrates that while the local sensitivity is preserved, the tuning overhead required to compensate for high-density disorder becomes a significant bottleneck for large-scale device operation.

IV. UNCOVERING THE STRUCTURE OF DISORDER

Having established the impact of disorder on key operational parameters, we now move beyond analyzing marginal distributions and uncover the underlying structure of the device-to-device variability. The correlations between parameters visualized in the corner plot in Fig. 6, contain crucial information about the common physical origin of the fluctuations. In this section, we

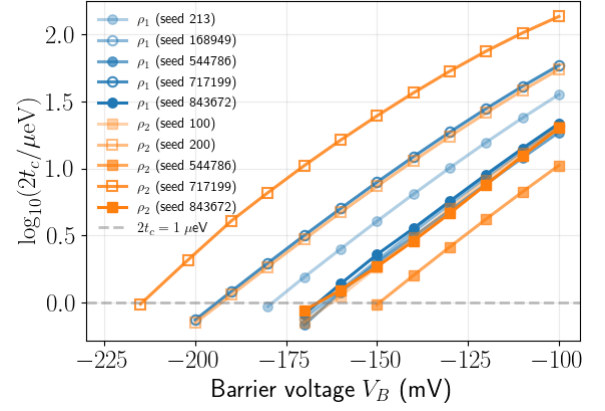


FIG. 5. Tunability of the tunnel coupling (t_c) as a function of barrier voltage (V_B) for low ($\rho_1 = 5 \times 10^9 \text{ cm}^{-2}$) and high ($\rho_2 = 5 \times 10^{10} \text{ cm}^{-2}$) disorder densities. The linear relationship in the logarithmic scale demonstrates that the sensitivity $\partial \log_{10}(2t_c)/\partial V_B$ is uniform across all disorder realizations (seeds) and independent of the total charge density. Crucially, the larger absolute spread of t_c at high densities (ρ_2) implies that a significantly wider sweep of V_B is required to bring an ensemble of disordered devices to a common low- t_c operating point, such as the $1 \mu\text{eV}$ threshold (horizontal dashed line). This highlights the increased “tuning overhead” imposed by electrostatic disorder in the high-density regime.

first construct a predictive model of the disordered device. We then use PCA to deconstruct the DQD parameter randomness due to disorder into its most physically meaningful components.

A. The Predictive Model

To capture the full statistical behavior of the DQD, we model the joint distribution of the parameter vector $\mathbf{X} = [d, \log_{10}(2t_c), L_x, \Delta L_z, F_z, \Delta F_z, \epsilon]^T$ as a multivariate normal distribution. The model is fully specified by the mean vector $\boldsymbol{\mu}$ and the covariance matrix $\boldsymbol{\Sigma}$, which are estimated directly from our ensemble of simulated devices:

$$P(\mathbf{X}) = \frac{1}{\sqrt{(2\pi)^k |\boldsymbol{\Sigma}|}} \exp \left(-\frac{1}{2} (\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \right), \quad (4)$$

$$\boldsymbol{\mu} = \langle \mathbf{X} \rangle, \quad \boldsymbol{\Sigma} = \langle (\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T \rangle. \quad (5)$$

This parameterized distribution, $P(\mathbf{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$, serves as a predictive statistical model, capable of generating realistic artificial device data by sampling from multivariate normal distribution.

To validate this model, we generate a synthetic dataset by sampling from the multivariate normal distribution in the PCA space and compare it to the original data from our full physical simulations. In Fig. 6, a corner plot overlaying the two datasets shows good agreement in the 1D marginals and the 2D correlations. To better capture the

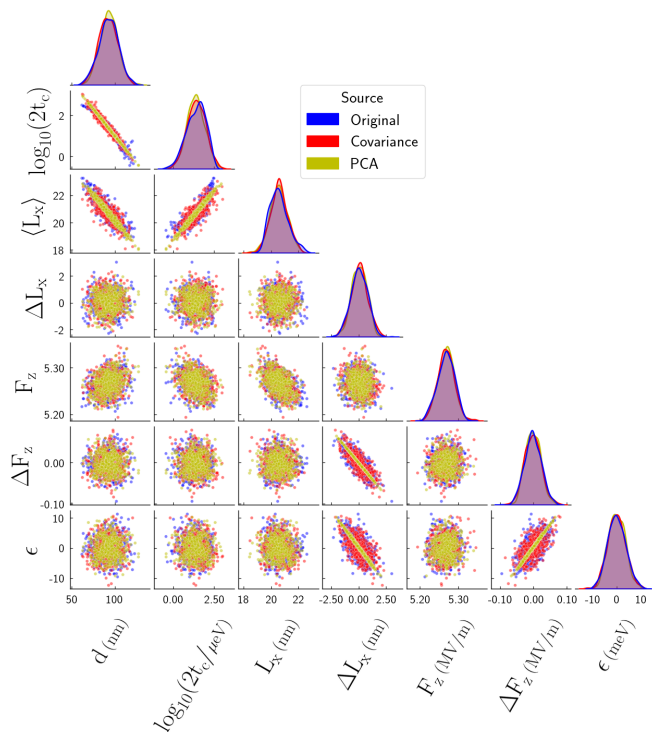


FIG. 6. **Model validation via synthetic sampling.** A scatter plot matrix showing pairwise correlations between key DQD parameters. Comparison of (i) a multivariate normal model (red) and (ii) a reduced principal-component model using the first three components (yellow) against the reference dataset from full simulations (blue). The diagonal cells show overlays of the three 1D marginals, demonstrating close agreement with the original data. All three data (original and reconstructed) are obtained for density $\rho = 5 \times 10^{10} \text{ cm}^{-2}$.

exponential scaling of the tunnel coupling with interdot distance and barrier height, we apply a log-transform to t_c prior to PCA. This improves the reproduction of the t_c marginal distribution and avoids unphysical negative values in the generated samples. Overall, the statistical model provides a high-fidelity representation of the device ensemble.

B. Principal Component Analysis

While the covariance matrix Σ contains all information about the correlations, its structure is not immediately interpretable. To identify the dominant, independent axes of variation, we perform principal component analysis (PCA) on the dimensionless correlation matrix,

$$C_{ij} = \frac{\Sigma_{ij}}{\sqrt{\Sigma_{ii}\Sigma_{jj}}}. \quad (6)$$

By solving the eigenvalue problem $\mathbf{C}\mathbf{d}_i = \lambda_i\mathbf{d}_i$, we obtain the eigenvectors $\mathbf{d}_i$ (the principal components) and their corresponding eigenvalues λ_i , which can be interpreted as the variance captured by each component [43].

The eigenvalue spectrum is shown in Fig. 7 (a). The variance is heavily concentrated in the first few components, with the first three PCs accounting for over 80% of the total variability. The remaining components fall within the random region predicted by the Marchenko-Pastur law [67]. As a result we assume any realization of disorder can be effectively approximated as a linear combination of just these three principal components.

To prove the point, in Fig. 6 we have also plotted the artificial data generated by sampling the distribution only along the first three principal components. The close agreement with the full dataset confirms that the essential correlated structure shared by both the real and artificial data corresponds to this three-dimensional subspace defined by the dominant principal components. By restricting the model to these top components, we explicitly discard the low-variance noise. This explains why the projected data appears cleaner than the full simulation set, as it retains the dominant physical correlations while filtering out minor fluctuations.

C. Physical Interpretation of Disorder Modes

We can understand the physical nature of these dominant modes by inspecting their composition, i.e. the elements of the eigenvectors, which are shown in Fig. 7 (b). Each component represents a specific, correlated way in which the DQD potential is deformed by the charge disorder, as illustrated schematically in Fig. 7 (c).

To be precise, in the first mode the disorder-induced increase of d is correlated with decrease of $2t_c$, decrease of confinement length x direction for both dots, and increase in the F_z electric field for both dots. This represents a symmetric squeezing or stretching of the DQD potential. An excess of negative charge located between the dots pushes the dots apart, leading to a decrease of $2t_c$.

The second mode is dominated by the fluctuations of energy detuning ϵ and the differential electric field ΔF_z . This corresponds to an asymmetric tilting of the double-well potential, which directly affects the detuning. This mode is primarily caused by excess charges located closer to one dot than the other.

The third significant mode is almost entirely composed of the average vertical field F_z . This represents a common-mode shift in the potential of both dots relative to the central barrier. It is distinct from PC_1 , as it has a weaker effect on the lateral confinement, and primarily reflects a change in the average vertical electric field, which is important for valley splitting [68, 69].

V. APPLICATION: DISORDER-AWARE CONTROLLABILITY

We now apply PCA to quantify the system controllability. We define controllability as the ability to mitigate

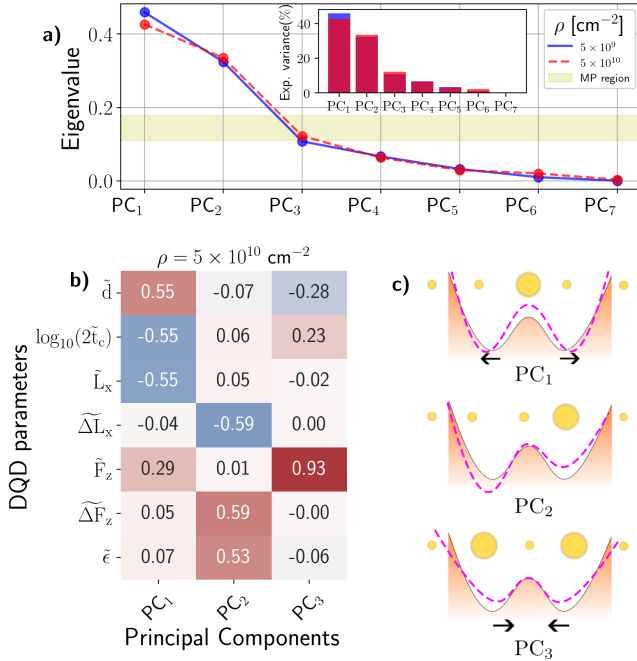


FIG. 7. **Parameter Correlations from disorder.** (a) The eigenvalues corresponding to each PC for two densities, $5 \times 10^9 \text{ cm}^{-2}$ (blue) and $5 \times 10^{10} \text{ cm}^{-2}$ (red). Inset: Corresponding explained variance of each PC. (b) Eigenvectors of the first three PCs, consistent with the schematic in (c). The datasets are chosen for fixed plunger gate $V_{L/R} = 0.35 \text{ V}$ and barrier gate $V_B = -0.1 \text{ V}$. In (c), small yellow dots represent average density of defect charges, while the large dot indicates the local excess (or suppression) of number of defect charges comparable to the average that responsible for the specific mode.

the effects of disorder by tuning the gate voltages. Specifically, we investigate whether a more scalable plunger-only control scheme [46] can mitigate disorder as effectively as the recently more common approach that uses both plunger and barrier gates.

A. Control Modes

First, we identify the control modes: the primary directions in the system's parameter space that can be accessed by tuning gate voltages. To do this, we generate a control dataset in the absence of disorder. Starting from a set of initial parameters, we sweep the gate voltages across their operational ranges: the barrier gate $V_B \in (-105, -95) \text{ mV}$, the average plunger voltage $V_P = (V_L + V_R)/2 \in (330, 380) \text{ mV}$, and the plunger voltage difference $\Delta V = V_L - V_R \in (0, 20) \text{ mV}$. The ΔV sweep is kept asymmetric ($\Delta V \geq 0$) to avoid introducing additional nonlinearities. This process yields a matrix of system parameters, $\mathbf{Y}_{3g} = \mathbf{Y}(V_P, \Delta V, V_B)$, where each row corresponds to a specific gate voltage configuration. We compare here the full 3-gate control with a plunger-only (2-gate) control scheme. For the latter, the

barrier gate is held fixed, yielding the parameter matrix $\mathbf{Y}_{2g} = \mathbf{Y}(V_P, \Delta V, V_B = \text{const})$.

We standardize both datasets to create a matrix $\tilde{\mathbf{Y}}$, where each column has a mean of zero and a standard deviation of one. PCA is then performed on $\tilde{\mathbf{Y}}$ to find the principal control modes, an orthonormal basis of vectors which we denote as $\mathbf{c}_j$. The top panel of Figure 8(a) shows the composition of these modes in the parameter space. Next, we determine the direction in the gate voltage space, β_j^c , corresponding to each principal mode using linear regression. Specifically, each β_j^c is found by solving the following least-squares problem:

$$\beta_j^c = \underset{\beta}{\text{argmin}} \sum_i [(\tilde{\mathbf{y}}_i \cdot \mathbf{c}_j) - (\Delta \mathbf{V}_i \cdot \beta)]^2, \quad (7)$$

Here, $\Delta \mathbf{V}_i = \mathbf{V}_i - \langle \mathbf{V} \rangle$ is the gate voltage vector for data point i centered around the dataset's mean voltage $\langle \mathbf{V} \rangle$, and $\tilde{\mathbf{y}}_i$ is the standardized system parameter vector. The resulting unit vectors in voltage space, $\beta_j^c / \|\beta_j^c\|$, define the principal control directions and are given in the bottom panel of Fig. 8(a). See Appendix C for more details.

As shown in Fig. 8(a) and (b), the number of statistically relevant control modes match the number of independent gate voltages, resulting in two for plunger-only control and three for 3-gate control. This one-to-one correspondence indicates that the system's response to gate voltages is approximately linear. It also suggests that the 2-gate control spans a two-dimensional manifold in the parameter space, which may be insufficient to compensate for higher-dimensional effects of disorder.

By definition, the voltage-space vectors β_j^c represent the optimal strategies for inducing maximum variation along the principal modes of the control data. For most control modes, this involves the simultaneous change of all available parameters. A notable exception is the third mode in the 3-gate scheme, which primarily creates strong, symmetric variations in the z-component of the electric field at the dot locations, leaving other system parameters largely unaffected.

B. Controllability

We now quantitatively check to what extent these control capabilities can counteract the device variability caused by the disorder. The effects of disorder are captured by a set of PCA vectors, the disorder modes $\{\mathbf{d}_j\}$, demonstrated in Fig. 7. We restrict our analysis to the first three disorder modes, which capture over 80% of the total variance.

To illustrate how well the control space covers the disorder space, we project the control data points onto the subspace spanned by the first three disorder modes calculated for two densities. The results, shown in Fig. 9, compare the projected 2-gate (red) and 3-gate (yellow) control data against the projected disorder data (blue). It is evident that the 3-gate control spans all three relevant dimensions of the disorder space, while the 2-gate

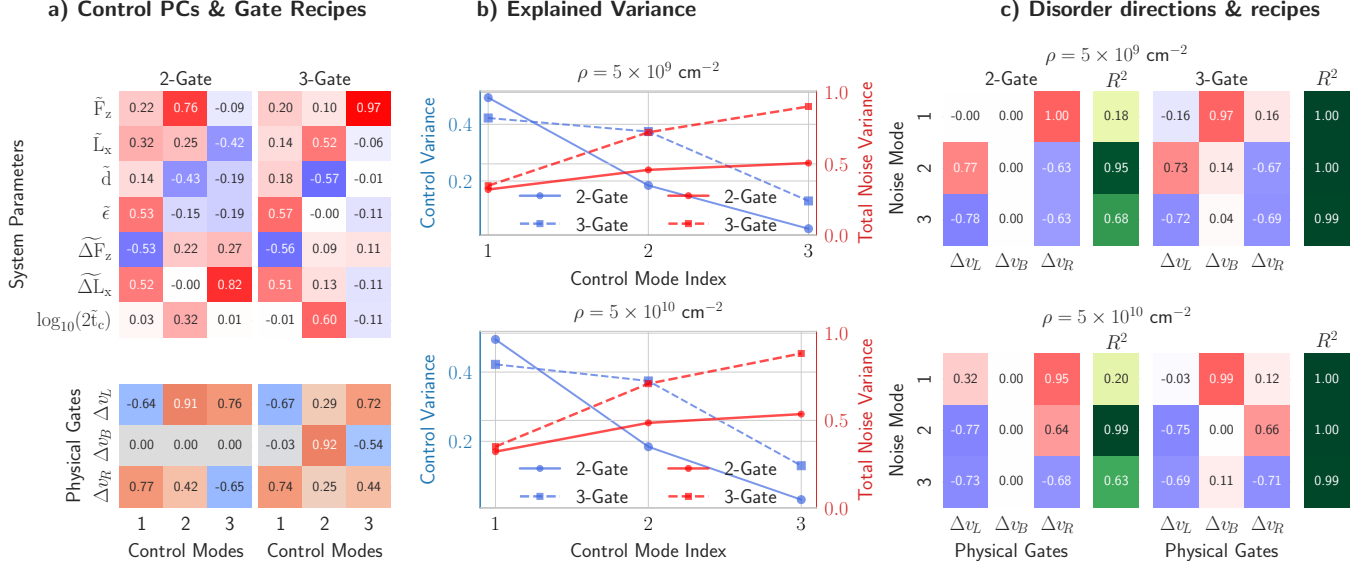


FIG. 8. **Comparison of two- and three-gate control schemes.** (a) Control modes and their voltage directions. Top: principal control modes $\mathbf{c}_j$ obtained by applying PCA to the standardized control dataset $\tilde{\mathbf{Y}}$ generated by sweeping gate voltages in the absence of disorder (2-gate: $(V_P, \Delta V)$ at fixed V_B ; 3-gate: $(V_P, \Delta V, V_B)$). Bottom: corresponding control directions in gate-voltage space, β_j^c , extracted by least-squares regression between mode amplitudes $(\tilde{\mathbf{y}}_i \cdot \mathbf{c}_j)$ and centered voltages $(\Delta \mathbf{V}_i \cdot \beta)$ [Eq. (7)]. (b) Controllability. Fraction of disorder variance explained by the first K control modes, η_K [Eq. (8)], where $\tilde{\mathbf{X}}$ is the standardized disorder dataset and N is the number of DQD parameters. Three-gate control reaches $\eta_K \gtrsim 0.9$ for $K = 3$, whereas plunger-only control saturates near $\eta_K \approx 0.5$ for $K = 2$. (c) Voltage recipes for compensating disorder modes. For each disorder principal component $\mathbf{d}_j$ ($j = 1, 2, 3$), the recipe β_j^d gives the optimal gate-voltage direction that best cancels that disorder mode; R_j^2 quantifies the quality of the linear compensation. Results are shown for (i) $\rho = 5 \times 10^9 \text{ cm}^{-2}$ and (ii) $\rho = 5 \times 10^{10} \text{ cm}^{-2}$. The disorder data used in this figure are post-tuning; a pre-/post-tuning comparison is provided in Fig. 12.

control is confined to an approximately two-dimensional subspace. In particular, the plunger-only control struggles to compensate for the first disorder mode, as the plane spanned by the red points is nearly orthogonal to the first disorder axis (disorder PC₁). The distinct “stripes” in the projected control data arise from the discrete voltage stepping used to map the control space. Successfully extracting the principal control axes from such sparse data reflects a practical strength of the PCA framework, demonstrating its robustness and direct applicability to time-constrained experimental calibrations.

Figure 8 (b) quantifies this by plotting the controllability (right axis), η_K , defined as the fraction of disorder variance explained by the first K control modes:

$$\eta_K = \frac{\sum_{j=1}^K \text{Var}(\tilde{\mathbf{X}} \cdot \mathbf{c}_j)}{N}. \quad (8)$$

Here, $\tilde{\mathbf{X}}$ is the standardized disorder dataset and N is the number of parameters. While the 3-gate control (red dashed line) can account for over 90% of the disorder variance, the plunger-only control (red solid line) corresponds to only about 50%, with most of that captured by its first control mode alone. This means that the control manifold spanned by the 3-gate scheme strongly overlaps with the directions of displacement caused by the disorder, whereas the directions spanned by 2-gate control

scheme is fundamentally unable to reach nearly 50% of the correlated disorder variations.

C. Optimal Compensation Recipes

Finally, we derive explicit gate voltage recipes to compensate for each disorder mode. We begin by decomposing each disorder mode $\mathbf{d}_j$ onto the basis of control modes $\mathbf{c}_i$:

$$\mathbf{d}_j \approx \sum_i \alpha_{ij} \mathbf{c}_i \quad (9)$$

The coefficients $\alpha_{ij} = (\mathbf{d}_j \cdot \mathbf{c}_i) / (\mathbf{c}_i \cdot \mathbf{c}_i)$ quantify how much each control mode $\mathbf{c}_i$ contributes to the disorder mode $\mathbf{d}_j$. The quality of this reconstruction is measured by the coefficient of determination, R_j^2 , defined as

$$R_j^2 = 1 - \frac{\|\mathbf{d}_j - \sum_i \alpha_{ij} \mathbf{c}_i\|^2}{\|\mathbf{d}_j\|^2}, \quad (10)$$

where a value close to 1 indicates that the disorder’s effects can be controlled.

An explicit gate recipe to counteract the j -th disorder mode is then the corresponding linear combination of the

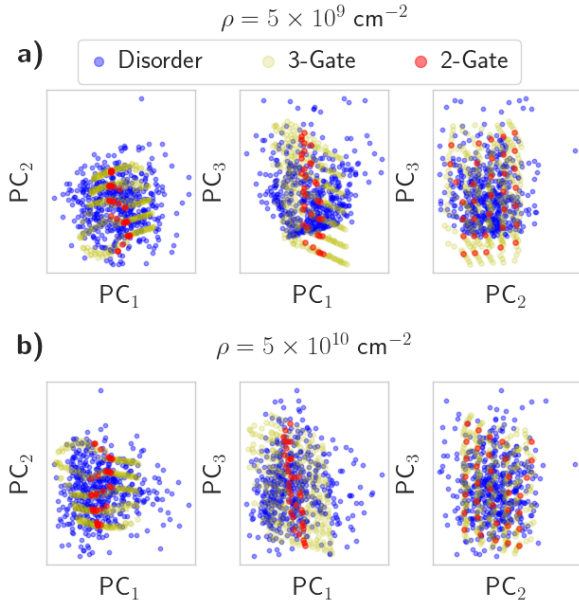


FIG. 9. Projected two-gate (red) and three-gate (yellow) normalized control data $\tilde{Y}$ compared with normalized disorder data $\tilde{X}$ (blue) for two densities 5×10^9 and 5×10^{10} , showing that three-gate control spans all relevant directions in full disorder space, while two-gate control is confined to an approximately two-dimensional subspace.

control mode recipes, β_i^c :

$$\beta_j^d = \sum_i \alpha_{ij} \beta_i^c \quad (11)$$

Figure 8(c) shows the derived voltage recipes, $\beta_j^d/|\beta_j^d| = (\Delta v_{Lj}^d, \Delta v_{Bj}^d, \Delta v_{Rj}^d)$, and their corresponding effectiveness, as quantified by the R_j^2 coefficient. The analysis quantitatively confirms the limitations of plunger-only control, reflected in the low R^2 value of 0.20 for the first disorder mode, which accounts for nearly half of the total variance. In contrast, the 3-gate control can effectively compensate for all three disorder modes, with R^2 values approaching 1.

Crucially, the derived voltage recipes are physically intuitive. While both control schemes can compensate for the second disorder mode (primarily a detuning shift) and the third mode (a common-mode shift), their responses to the first disorder mode are very different. For 3-gate control, this strong disorder in the inter-dot potential can be effectively compensated by changing the barrier gate voltage, an action unavailable in the plunger-only scheme.

Our framework demonstrates robustness not only against the overall disorder strength (various densities ρ) but also against shifts in the pre-set gate voltages ($V_{L/R}$ and V_B) that define the initial operating point and the parameters of the dot in the disorder-free structure. While the shapes of marginal distributions of the key DQD parameter do depend on the choice of these

voltages, the PCA results, i.e. the structure of correlations between the DQD parameters, and the conclusions concerning the controllability of DQDs, are robust to the choice of them. This is illustrated in Appendix A, confirming the wide applicability of our findings to Si/SiGe DQDs.

We have also repeated in Appendix B the above data analysis procedure to compare the controllability before and after the pre-tuning procedure, the detuning correction required to obtain a symmetric DQD. This dataset was post-selected based on tunability success rate conditions to avoid pathological cases, such as a single elongated dot. Crucially, the controllability results remain consistent with the finely-tuned case, confirming the model's predictive power.

To assess the practical accuracy of Eq. (11), we applied the corresponding voltage recipes to specific disorder realizations and recomputed the device parameters at the corrected solution voltages. For each realization, the disorder-induced shift was projected onto the PCA basis, and the resulting coefficients were used to construct the compensating gate-voltage shift predicted by $\Delta V = -\kappa \sum_i \gamma_i^{c/d} \beta_i^{c/d}$, where $\gamma_i^{c/d}$ represents the magnitude of the error for a specific mode i , and $\beta_i^{c/d}$ represents the corresponding gate-voltage recipe. This approach implements the intuitive rule to move against the measured PCA displacement, where the resulting update scales in Volts via the parameter κ to map the dimensionless PCA coordinates back to the physical gate-voltage domain.

This single-device validation shows that the recipe can effectively drive the system back toward the symmetric operating point for the dominant disorder modes, confirming the operational value of the linear PCA framework. However, the procedure also exposes its physical limits: the simultaneous compensation of all disorder principal components (PCs) requires careful execution. Specifically, attempts to compensate for disorder-induced displacements along the PC_1 and PC_2 directions introduce an unintended shift along PC_3 due to small system non-linearities. Combined with the low statistical significance of PC_3 , this makes the third direction highly susceptible to cross-talk. This issue can be mitigated either by performing sequential corrections along individual directions, or by simply neglecting the PC_3 correction if shifts in the global vertical electric field F_z (the main constituent of PC_3) do not significantly impact qubit operation. A detailed derivation of this procedure, along with specific examples of individual realizations and an evidence of this cross-talk, is presented in Appendix C.

VI. DISCUSSION AND CONCLUSION

Our analysis demonstrates that charge disorder in Si/SiGe double quantum dots, while a source of significant device-to-device variability, does not lead to mutually independent fluctuations of DQD parameters. Instead, it induces highly structured variations of these pa-

rameters along specific modes. These modes can be identified with certain real-space arrangements of charges in the oxide that lead to correlated fluctuations in the DQD parameters.

Beyond this specific insight, our work establishes the PCA as a powerful framework for analysis of variability of properties of semiconductor nanostructures that host spin qubits. Based on spectral analysis of the normalized covariance matrix, it provides a systematic tool for analyzing multivariate data. In particular, PCA allowed us to decompose the disorder fluctuations into three dominant modes: a symmetric squeezing/stretching of the DQD potential, an asymmetric tilting of the double-well potential, and a common-mode shift in the potential of both dots relative to the central barrier. These three modes account for more than 80% of the total variance in the DQD parameters and allow for a clear physical interpretation of the effect of the disorder.

By quantifying these correlations, we have constructed a multivariate Gaussian model, which is fully specified by the mean and covariance matrix. In this way, we have developed a predictive model that can be used to generate realistic artificial data without the need for computationally expensive electrostatic simulations, thereby facilitating the development of advanced tuning algorithms.

By characterizing the amplitude of the fluctuations, we have also analyzed how the variability changes as a function of charged defect density ρ over an order-of-magnitude wide range of ρ . We have confirmed that reproducibility, quantified by the device fulfilling the chosen here tunability success conditions, drops as disorder amplitude increases. We have analyzed the marginal distributions of the key DQD parameters, and found that while many of them are approximately Gaussian, the tunnel coupling specifically shows non-Gaussian features; however, by applying a log-transform to t_c , we recover a Gaussian-like distribution and ensure physical consistency in our model. We have related the variability of the tunnel coupling to fluctuations in the inter-dot distance, which we have found to be strongly correlated with the barrier height. This observation suggests that a measurement of t_c can serve as a direct probe of the inter-dot distance, with possible verification through measuring spin splittings of two single-spin qubits in the two dots in presence of a magnetic field gradient [3–5]. We have also analyzed the implications of this tunnel coupling variability for charge shuttling, finding that for a charge density above $\rho = 5 \times 10^9 \text{ cm}^{-2}$ ($\rho = 5 \times 10^{10} \text{ cm}^{-2}$), at least 6% (25%) of the dot pairs in a chain would require resource-intensive, individual barrier gate tuning.

As a novel in context of control of quantum dots application of PCA, we have used it to quantify the limitations of plunger-only control. We have compared the three-mode PCA decomposition of the disorder space with the PCA decomposition of the control space, which was obtained by sweeping the gate voltages in the absence of disorder. A comparison of the manifolds spanned by the

most important PCs allowed us to visualize the overlap between the control and disorder spaces, and to quantify system controllability, defined as the ability to mitigate the effects of disorder by tuning gate voltages. We have found that while 3-gate control can explain over 90% of the disorder variance, plunger-only control explains only about 50%.

For practical purpose of mitigating these disorder modes, we have derived explicit gate voltage recipes to compensate for each one. The results confirm that plunger-only control is severely limited in its ability to counteract the first, dominant disorder mode, which accounts for nearly half of the total variance of DQD parameters. This stands in stark contrast to three-gate control, which can effectively compensate for all three modes with recipes that directly match their physical interpretation. We empirically validated these recipes on individual disorder realizations, confirming their ability to tune the system back to the disorder-averaged point, although this testing also revealed that small system nonlinearities can introduce cross-talk when simultaneously compensating for all modes, an effect that is significant only for the less relevant modes. Furthermore we identified limitation of two gate control. While plunger-only control can address the second mode (a detuning shift) and the third mode (a common-mode vertical shift), it struggles with the first mode. The symmetry of the two plunger gates makes it difficult to compensate for changes in d (and hence in t_c) without simultaneously altering the potential symmetry and introducing unwanted changes in ϵ , ΔL_x , and ΔF_z .

This suggests that while plunger-only control is a more scalable approach, it may require more sophisticated strategies. For instance, it might be necessary to introduce additional control knobs by relying on cross-talk from neighboring plunger gates. Alternatively, one could operate in a non-linear regime where the manifold spanned by the two control modes is a curved surface that could potentially cover more of the disorder space. Of course, one can try to correct for a specific parameter, for instance t_c , by tuning the plunger gates in a symmetric way. However, our analysis suggests that such an approach will lead to a simultaneous change in the electric field F_z while failing to correct the change in d . The former might adversely modify the valley physics in Si/SiGe qubits and the electric control of spin qubits in all platforms, while the latter leaves a primary issue for two-qubit gates and the operation of singlet-triplet qubits unresolved.

Finally, we acknowledge that charge disorder is not the sole factor limiting device tunability. In experimental settings, fabrication-induced imperfections and geometric variability, such as layer misalignment or fluctuations in gate dimensions, constitute an equally significant source of parameter spread. While our study focuses on isolating electrostatic disorder to rigorously quantify the impact of electrostatic disorder, these structural variations are critical bottlenecks for the large-scale integration of spin qubits. Importantly, the statistical framework es-

established here is generalizable: the PCA approach can be readily applied to geometric datasets to decompose fabrication-induced variability into a few dominant, tunable control modes, providing a path toward hardware-aware compensation for structural defects. It should be noted that since we focus on electrostatic disorder only, we neglect the randomness of valley coupling values (and thus of valley splitting values) in the two dots resulting from the roughness of the Si/SiGe interface and the inter-diffusion of Ge atoms [33, 44, 50, 70, 71]. Unequal values of complex valley coupling parameter in the two dots are known to affect the tunnel coupling between the valley and orbital ground states in the two dots [72–74]. Our neglect of this effect means that the results presented in this paper for the randomness of t_c due to electrostatic disorder apply to the case of atomic disorder at Si/SiGe interface resulting in the “deterministic” regime of valley couplings [33, 75]. In the case of currently often encountered Si/SiGe dots with valley coupling in the “disordered” regime [33, 76], the tunnel coupling t_c calculated here is related to the inter- and intra-valley tunnel couplings $t_{\pm}$ by $t_c^2 = |t_+|^2 + |t_-|^2$ [72–74]. The atomic disorder at Si/SiGe interface also affects any properties of Si/SiGe spin qubits that are connected with the effective spin-orbit coupling in a given dot, such as the electron g -factors [77, 78] and spin-valley couplings [75, 76, 79]. While the knowledge of variability of parameters such as confinement length, dot ellipticity, and electric fields in the z direction that affect the overlap of the electron wavefunction with the interface, will be relevant for full characterization of variability of g -factors and spin-valley couplings, including the atomistic disorder at the interface will be crucial for these properties.

Concluding, we have analyzed the device-to-device variability in Si/SiGe double quantum dots arising from electrostatic disorder and developed a predictive statistical model to generate reliable artificial data for tuning algorithm development. Using the Principal Component Analysis on a large simulated dataset, we have found that the resulting parameter fluctuations were taking place along a few directions in parameter space. We have identified three such fluctuation modes, with the most significant one being caused by fluctuations in number of defect charges localized in-between the dots. This mode leads to correlated fluctuations in the inter-dot distance and barrier height that can fully explain the strong variation in the tunnel coupling. By quantifying the system controllability, we have shown that such a mode cannot be fully corrected by a plunger-only control scheme, which lacks the necessary barrier control. Our work establishes PCA as a powerful framework for understanding and mitigating disorder in quantum dot devices, and highlights the importance of barrier control for scalable qubit operation.

CODE AND DATA AVAILABILITY

The data and source code required to reproduce the findings of this study are available on Zenodo [80] and GitHub [81], respectively.

ACKNOWLEDGMENTS

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Appendix A: PCA and fluctuations for different pre-set gate voltages ($V_{L/R} = 0.19$ V and $V_B = -0.06$ V)

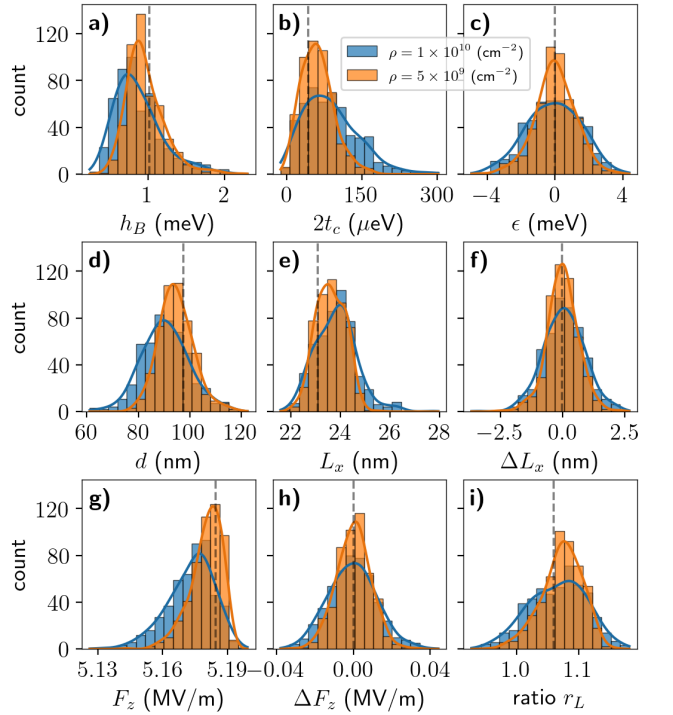


FIG. 10. Marginal distributions of the key DQD parameters as function of charge density ρ . The corresponding ensemble-averaged value of each DQD parameter are summarized in Table II. The vertical dashed line indicates the nominal defect-free DQD baseline (set by $V_0 = 0.19$ V and $V_B = -0.06$ V). The key parameters for this baseline are: interdot distance $d_0 = 97$ nm, barrier height $h_B = 1.01$ meV, and tunneling gap $2t_{c,0} \approx 42$ μ eV. The dots are elliptical with lateral dimensions $L_{x,0} \approx 24$ nm and $L_{y,0} \approx 22$ nm (ratio $r_D = 1.09$) with average electric field $F_z \approx 5.18$ MV/m.

TABLE II. Statistical summary of parameters, including mean, standard deviation (std), and the coefficient of variation (CV) for two densities $\rho_1 = 5 \times 10^9$ and $\rho_2 = 1 \times 10^{10}$ cm^{-2} . The datasets are chosen for fixed plunger gate $V_{L/R} = 0.19$ V and barrier gate $V_B = -0.06$ V.

Parameter		Mean (μ)	Std (σ)	CV ($\sigma/ \mu $)
d (nm)	ρ_1	94.67	6.67	0.07
	ρ_2	90.54	9.01	0.09
h_B (meV)	ρ_1	0.96	0.25	0.26
	ρ_2	0.87	0.31	0.35
ϵ (meV)	ρ_1	0.02	1.17	—
	ρ_2	-0.09	1.58	—
F_z (MV/m)	ρ_1	5.18	0.007	0.001
	ρ_2	5.17	0.01	0.002
ΔF_z (MV/m)	ρ_1	$< 10^{-4}$	0.008	—
	ρ_2	$< 10^{-4}$	0.012	—
ΔL_x (nm)	ρ_1	$< 10^{-3}$	0.63	—
	ρ_2	0.05	0.81	—
L_x (nm)	ρ_1	23.60	0.58	0.02
	ρ_2	23.81	0.82	0.03
$2t_c$ (μeV)	ρ_1	60.96	31.70	0.52
	ρ_2	87.01	53.33	0.60
$r_L, [r_R]$	ρ_1	1.07, [1.07]	0.03, [0.03]	0.03, [0.03]
	ρ_2	1.06, [1.06]	0.045, [0.046]	0.042, [0.043]

Note. CV is not informative when the mean is close to zero or changes sign (e.g., ϵ , ΔF_z , ΔL_x); the large values reflect the instability of $\sigma/|\mu|$. For these quantities, variability is better characterized by σ . Non-zero means (μ) likely arise from finite-size and numerical-accuracy effects.

To further test the robustness of our conclusions and the applicability of the PCA approach across different regimes, we performed the same analysis using for another set pre-tuned values for the plunger voltages ($V_{L/R} = 0.19$ V) and the gate barrier voltage ($V_B = -0.06$ V). We perform this analysis to verify that the results for the PCA and controllability are not sensitive to the choice of voltages defining the shape and properties of DQD in a disorder-free structure, whereas the fluctuations of key DQD parameters depend on the gate voltages and charge densities. Let us note that with the voltages chosen here the double-well character of the electrostatic potential in absence of disorder is less pronounced than for parameters used in the main text, and as a result the DQD would vanish for many realizations of charge disorder for the highest $\rho = 5 \times 10^{10} \text{ cm}^{-2}$ considered in the paper. Consequently, we focus here on lower densities, $\rho_1 = 5 \times 10^9 \text{ cm}^{-2}$ and $\rho_2 = 1 \times 10^{10} \text{ cm}^{-2}$.

In Fig. 10 the corresponding marginal distributions of key parameters for two charge densities are shown. Similarly, while many of the parameters follow approximately Gaussian distributions, certain quantities—such as tunnel gap ($2t_c$), barrier height (h_B), and vertical electric field (F_z) again show significant non-Gaussian behavior.

For these two considered densities that differ by a factor of two, the broadening of the higher-density distribution relative to the lower one is less pronounced compared to the cases shown in the main text (see Fig. 3), where the densities there differ by an order of magnitude. Nevertheless, the difference remains observable. The details, including the mean, standard deviation, and coefficient of variation, are presented in Table II.

Crucially, the results of the PCA calculations for this new voltage setting are quantitatively very similar to those for the previous one, suggesting that the nature of these dominant modes is primarily related to the spatial distribution of defects rather than the initially pre-set gate voltage values. The resulting eigenvalue spectra and eigenvector compositions, presented in Fig. 11, show that the variance remains predominantly concentrated in the first few principal components, with the first three accounting for over 80% of the total variability—closely matching the behavior observed at the reference density in Fig. 7. Moreover, the spatial profiles of the dominant eigenmodes exhibit similar patterns of correlated potential deformation, reinforcing that the fundamental nature of disorder-induced fluctuations is largely insensitive to charge density within this range.

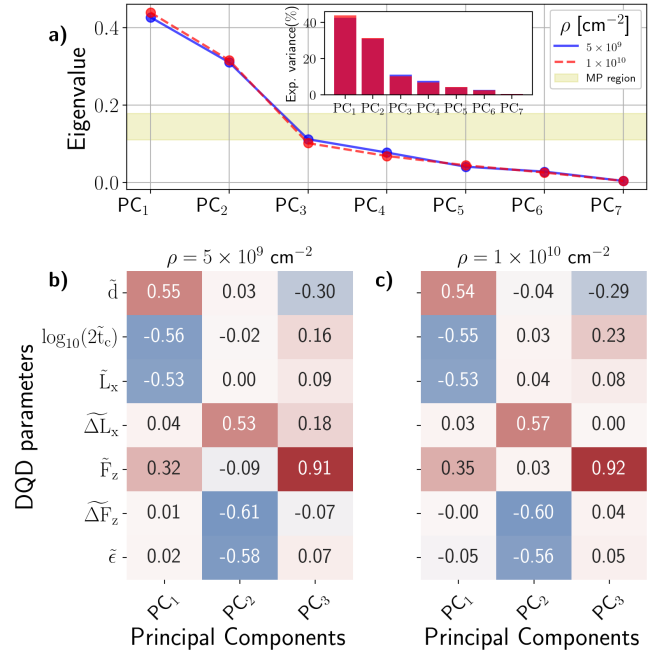


FIG. 11. **Parameter correlations from disorder.** (a) The eigenvalues corresponding to each PC for two densities, $5 \times 10^9 \text{ cm}^{-2}$ (blue) and $1 \times 10^{10} \text{ cm}^{-2}$ (red). Inset: Corresponding explained variance of each PC. (b)-(c) Eigenvectors of the first three PCs. The datasets are chosen for fixed plunger gate $V_{L/R} = 0.19$ V and barrier gate $V_B = -0.06$ V.

Appendix B: Controllability Results for DQD without Pre-tuning

Using the modified gate settings defined in Appendix A, we apply the same PCA-based approach to quantify the controllability for a dataset at a density of $\rho = 1 \times 10^{10} \text{ cm}^{-2}$. In this way we benchmark our approach in two ways. First, we have changed the tuning point to analyze the controllability of shallower dots at smaller charge densities. Second, we analyze two distinct scenarios: one dataset with the typical pre-tuning (detuning correction) applied, and one without any detuning correction. This benchmarking is crucial because the influence of charged defects strongly depends on their density ρ and the pre-set gate voltages, especially when working with shallower (less confined) dots.

In contrast to the main text, here we operate at shallower-dots conditions ($V_{L/R} = 0.19 \text{ V}$, $V_B = -0.06 \text{ V}$), which allow to observe DQDs for densities of $\rho = 5 \times 10^9 \text{ cm}^{-2}$ and $\rho = 1 \times 10^{10} \text{ cm}^{-2}$, but not at the higher $\rho = 5 \times 10^{10} \text{ cm}^{-2}$ considered in the main text. At these lower densities, the defects mainly introduce a modest shift and deformation in the electrostatic potential, which still allows the two distinct quantum dot minima to remain clearly visible for most disorder realizations. We highlight that at higher densities ($\rho = 5 \times 10^{10} \text{ cm}^{-2}$), the defect influence is strong enough to completely wash out one or both dots, often resulting in shallow or single-dot potentials, which are far from the desired operating point. Overcoming this issue requires significant adjustment of the gate voltages, often demanding stronger initial confinement - and this is what we have done in the main text. For shallower dots, we again rely on sweeping the gate voltages across their operational ranges: the barrier gate $V_B \in (-60, -50) \text{ mV}$, the average plunger voltage $V_P = (V_L + V_R)/2 \in (180, 210) \text{ mV}$, and the plunger voltage difference $\Delta V = V_L - V_R \in (0, 10) \text{ mV}$.

The controllability results using dataset after detuning correction shown in Fig. 12 are very similar to those discussed in Sec. V. This similarity is not surprising, given the almost identical control modes and PCA. The consistency of these results is preserved when examining the dataset before detuning correction (see bottom panel of Fig. 12 (d)). Here, the voltage recipes and their corresponding effectiveness (R^2) for the first three control modes are similar to those obtained for post-tuned dataset. Our analysis demonstrates that working in the regime in which both of the dots are well defined, while being detuned by a few meV, yields control analysis consistent with the one obtained at adjusted operating points.

Appendix C: Correction Methodology Using PCA Recipes

To compensate for device-to-device variability induced by random potential fluctuations (disorder), we em-

ploy a projection-based steering algorithm that operates directly in the three-dimensional gate-voltage space ($\Delta V_L, \Delta V_B, \Delta V_R$) while using a reduced coordinate system defined by principal component analysis (PCA). The central idea is to decompose disorder-induced parameter shifts into a low-dimensional PCA basis and to translate the resulting PCA coordinates into coordinated gate-voltage moves (“recipes”) that steer the device back toward a fixed operating point. In this work, the reference is defined as the mean of the successfully tuned ensemble, denoted by μ .

1. Disorder versus control manifold

In the following, the subscript $s \in \{c, d\}$ labels the PCA subspace used for the update: $s = c$ refers to the control PCA basis, while $s = d$ refers to the disorder PCA basis. All quantities carrying the subscript s are defined analogously, differing only in which PCA basis and scaling are employed. Since the control and disorder data are taken separately, we use $\mathbf{X}_d \equiv \mathbf{X} \in \mathbb{R}^{M \times N_d}$ to denote parameters sampled under N_d different realizations of the disorder, and $\mathbf{X}_c \equiv \mathbf{Y}$ to denote parameters recorded under N_c different voltage setups $\mathbf{V} \in \mathbb{R}^{3 \times N_c}$ in the absence of the disorder.

Let $\mathbf{x}_s \in \mathbb{R}^M$ be a single data point, i.e., a single row of $\mathbf{X}_s$. For each dataset, we use $\mu_s \in \mathbb{R}^M$ to denote the mean of each parameter and $\sigma_s \in \mathbb{R}^M$ to denote the standard deviation of each parameter. In our case, both $\mu_d \neq \mu_c$ and $\sigma_d \neq \sigma_c$, as the control data is gathered in the absence of disorder. For a chosen subspace $s \in \{c, d\}$, we define the diagonal scaling matrix $\mathbf{S}_s$ that effectively divides each parameter by its ensemble standard deviation $\sigma_{s,m}$, resulting in a standardized datapoint:

$$\tilde{\mathbf{x}}_s = \mathbf{S}_s (\mathbf{x}_s - \mu_s) \in \mathbb{R}^M. \quad (\text{C1})$$

Next, we perform the PCA in the chosen subspace s by an eigendecomposition of the correlation matrix. Let $\mathbf{e}_{s,i}$ be the top K_s eigenvectors of Σ_s (ordered by decreasing eigenvalue); stacking them as rows yields the projection matrix

$$\mathbf{D}_s = \begin{bmatrix} \mathbf{e}_{s,1}^\top \\ \vdots \\ \mathbf{e}_{s,K_s}^\top \end{bmatrix} \in \mathbb{R}^{K_s \times M}. \quad (\text{C2})$$

The device’s reduced-coordinate fingerprint in subspace s is then

$$\gamma^s = \mathbf{D}_s \tilde{\mathbf{x}}_d \in \mathbb{R}^{K_s}, \quad (\text{C3})$$

with γ_i^s denoting the projection of the disorder datapoint $\tilde{\mathbf{x}}_d$ into the K_s -dimensional space spanned by the most relevant PCs. The γ^s vector can be treated as coordinates in the manifold spanned by those PCs. One can also use them to compare the relevant directions of control and disorder data in the relevant PC space. For

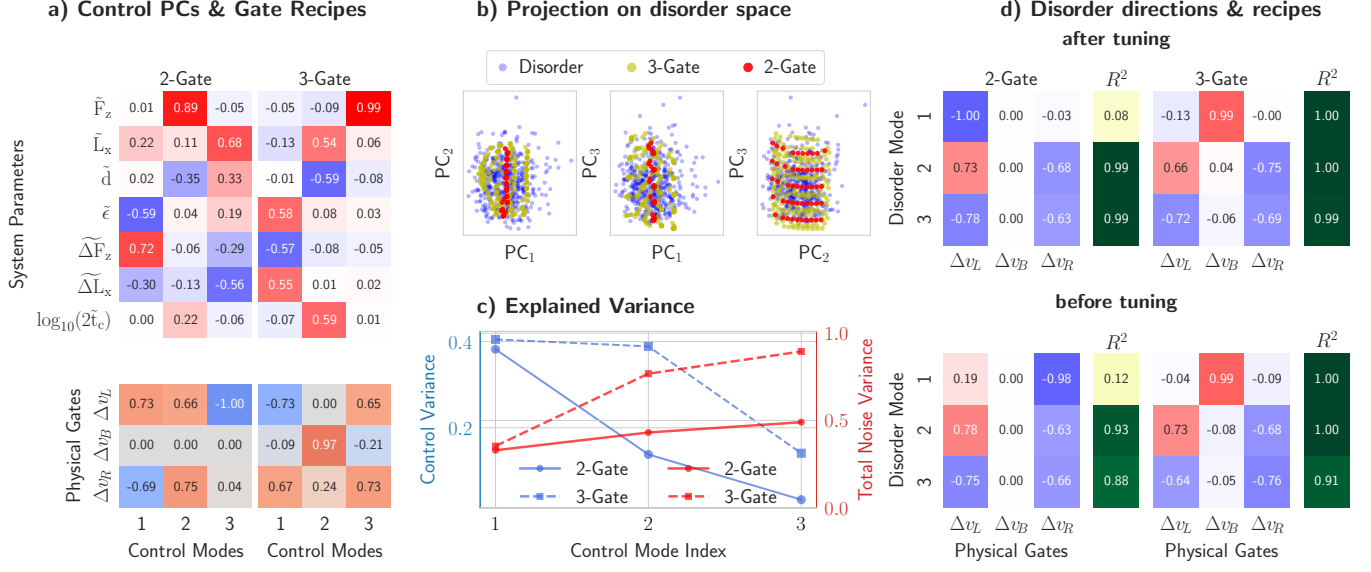


FIG. 12. **Comparison of two- and three-gate control schemes.** (a) Composition of control modes in gate-voltage space and determination of control directions, β_j^c , via least-squares regression. (b) Projected two-gate (red) and three-gate (yellow) control data compared with disorder data (blue), showing that three-gate control spans the full disorder space, while two-gate control is confined to an approximately two-dimensional subspace. (c) Controllability, η_K , representing the fraction of disorder variance explained by the K th control mode. The three-gate control (blue line) accounts for over 90% of the variance, whereas the plunger-only control (red line) captures about 50%. Post-tuned dataset used for (b) and (c). (d) Derived voltage recipes, β_j^d , and corresponding R_j^2 values for (i) post-tuning and (ii) pre-tuning datasets. The disorder data is chosen for density $\rho = 1 \times 10^{10} \text{ cm}^{-2}$. The datasets are chosen for fixed plunger gate $V_{L/R} = 0.19 \text{ V}$ and barrier gate $V_B = -0.06 \text{ V}$.

instance, in Fig. 9 we plot the separately normalized disorder and control datasets, before projecting them onto the same disorder PC manifold, i.e., $\gamma^d = \mathbf{D}_d \tilde{\mathbf{x}}_d$ for the disorder data, but $\gamma^{dc} = \mathbf{D}_d \tilde{\mathbf{x}}_c$ for illustrating control data.

2. Steering disorder-induced displacements

We now attempt to counteract these deviations by finding a voltage recipe that corrects a particular disorder-induced displacement $\tilde{\mathbf{x}}_d$.

To achieve this, we first determine how the physical gate voltages map onto the principal components of the control space. For the control dataset, let $\Delta \mathbf{V}_i = \mathbf{V}_i - \langle \mathbf{V} \rangle$ be the centered gate voltage vector applied to generate the i -th sample $\tilde{\mathbf{x}}_{c,i}$. The optimal gate-voltage direction β_j^c that drives the system along the j -th control mode $\mathbf{e}_{c,j}$ is extracted via ordinary least squares:

$$\beta_j^c = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{N_c} [(\tilde{\mathbf{x}}_{c,i} \cdot \mathbf{e}_{c,j}) - (\Delta \mathbf{V}_i \cdot \beta)]^2. \quad (\text{C4})$$

Next, we derive the explicit gate voltage recipes for the disorder modes by decomposing each disorder principal component $\mathbf{e}_{d,j}$ onto the basis of the control modes $\mathbf{e}_{c,k}$:

$$\mathbf{e}_{d,j} \approx \sum_k \alpha_{jk} \mathbf{e}_{c,k}, \quad (\text{C5})$$

where $\alpha_{jk} = (\mathbf{e}_{d,j} \cdot \mathbf{e}_{c,k}) / (\mathbf{e}_{c,k} \cdot \mathbf{e}_{c,k})$. The gate recipe to counteract the j -th disorder mode is then the corresponding linear combination of the control mode recipes, $\beta_j^d = \sum_k \alpha_{jk} \beta_k^c$.

To isolate the pure direction of the correction, we normalize these vectors to obtain unit-direction gate trajectories, $\hat{\beta}_i^s = \beta_i^s / \|\beta_i^s\|$. These normalized vectors populate the optimized gate-recipe matrix $\mathbf{B}_s \in \mathbb{R}^{K_s \times 3}$ for the chosen subspace $s \in \{c, d\}$. The matrix stores, row-by-row, these unit vectors:

$$\hat{\beta}_i^s = [\hat{\beta}_{i,L}^s \ \hat{\beta}_{i,B}^s \ \hat{\beta}_{i,R}^s] \in \mathbb{R}^3, \quad (\text{C6})$$

in (V_L, V_B, V_R) space associated with each PCA mode. The resulting dimensionless gate update is computed as the negative weighted sum:

$$\Delta \mathbf{v}_s = -\mathbf{B}_s^\top \gamma^s \in \mathbb{R}^3, \quad (\text{C7})$$

which implements the intuitive rule to “move against the measured PCA displacement.” The resulting dimensionless update $\Delta \mathbf{v}_s$ is mapped to a physical voltage change by a scalar step-size parameter κ . Here, κ serves as a global scaling factor with units of Volts, reflecting the conversion from standardized PCA coordinates back to the experimental gate-voltage domain:

$$\Delta \mathbf{V} = \kappa \Delta \mathbf{v}_s. \quad (\text{C8})$$

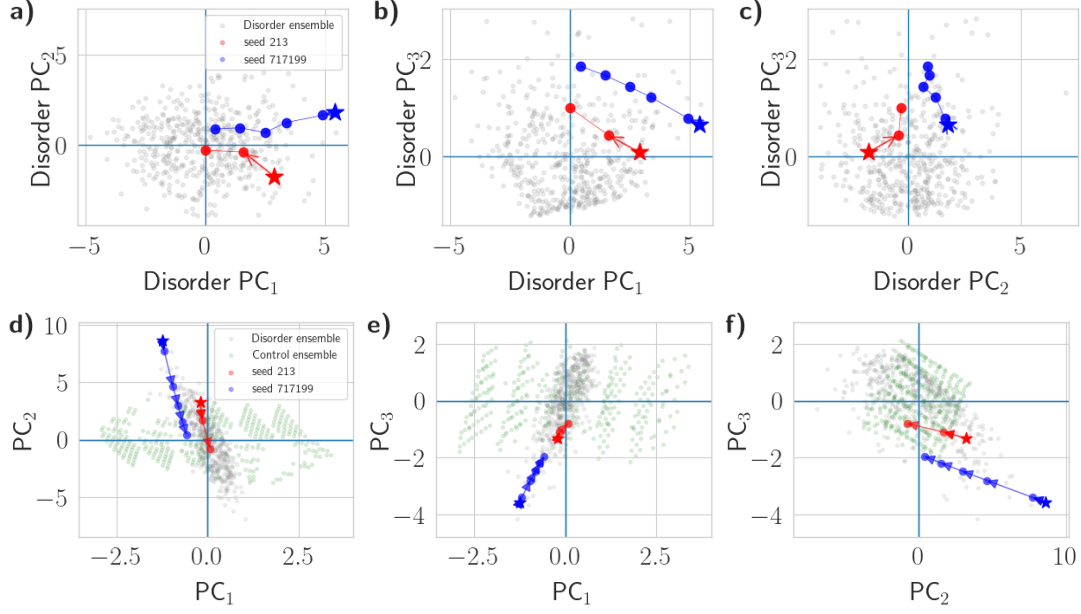


FIG. 13. Trajectory of two specific disorder realizations (seeds 213 and 717199 for $\rho = 5 \times 10^9 \text{ cm}^{-2}$) in the principal component (PC) space during the application of the corrective voltage recipes derived from Eq. (11). The top row shows the movement within the disorder PC space, while the bottom row displays the corresponding path in the control PC space. The background ensemble of devices is shown in light blue. The applied correction successfully steers the disordered devices toward the origin (the symmetric, unperturbed state) when projecting the parameters on PC_1 and PC_2 disorder modes, visualizing the effectiveness of the linear compensation. The compensation along the PC_3 disorder mode does not work well for reasons discussed in the text.

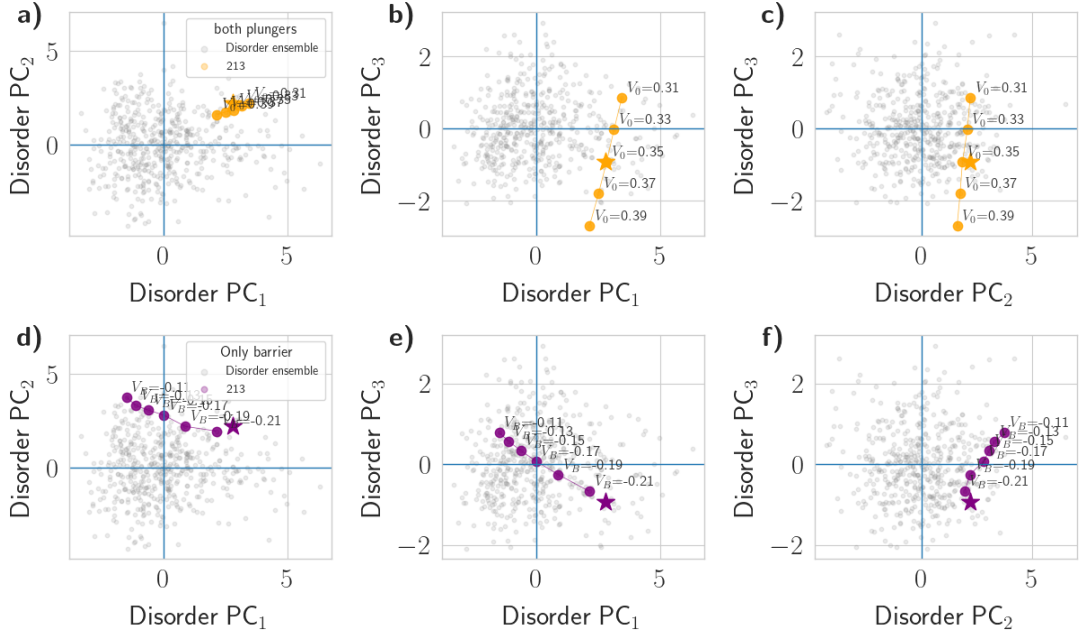


FIG. 14. Independent effect of specific gate sets on the device state in the disorder PC space for a single disorder realization (seed 213 for $\rho = 5 \times 10^{10} \text{ cm}^{-2}$). The top row demonstrates the system trajectory when iteratively adjusting only the plunger gates, while the bottom row shows the trajectory when adjusting only the barrier gates. This confirms the physical decoupling of the control space, illustrating how different physical gates couple to, and drive, distinct principal components of the disorder.

3. Empirical evidence

The effect of this steering on the system state is shown in Fig. 13. While this PCA-based correction effectively

brings the DQD parameters significantly closer to the background values, testing these specific devices allowed

us to identify and document several physical limitations. First, the linear correction exhibits residual nonlinearities, most noticeably in the average vertical field (F_z) and the interdot field difference (ΔF_z). Due to these higher-order effects, it is not possible to return an arbitrary realization exactly to the ensemble-average background point μ_d while matching all seven parameters simultaneously.

Second, the third principal component of the disorder lies near the threshold of statistical significance. Although its physical signature can be identified as a coordinated plunger-pushing mode, its smaller eigenvalue relative to PC_1 and PC_2 implies a relatively large sensitivity to uncorrelated noise and potential residual correlations with the more dominant directions. This is confirmed in

Fig. 14, where we show the effects of applying a specific ΔV to steer the parameters along a targeted direction. While a common-mode plunger sweep has the largest effect on PC_3 , as expected, a noticeable movement along the PC_1 direction can also be observed. Similarly, for a barrier gate sweep, which should primarily correspond to the PC_1 direction, the responses of PC_1 and PC_3 are comparable, confirming significant cross-talk between them. Furthermore, at sufficiently large barrier sweeps, movement along PC_2 is also observed, demonstrating the limitations of the linear response approximation.

Despite these higher-order effects, these results validate the PCA framework as a highly effective and physically intuitive method for locally steering the parameters of a disordered DQD toward a functional operating regime.

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